But irrespective of your point of view on the hypothetical question of teleporting a living being, scientists have now established that, through the wonders of quantum mechanics, *individual particles can be—and have been—teleported.*

Let's see how.

Quantum Entanglement and Quantum Teleportation

In 1997, a group of physicists led by Anton Zeilinger, then at the University of Innsbruck, and another group led by A. Francesco De Martini at the University of Rome,[2] each carried out the first successful teleportation of a single photon. In both experiments, an initial photon in a particular quantum state was teleported a short distance across a laboratory, but there is every reason to expect that the procedures would have worked equally well over any distance. Each group used a technique based on theoretical insights reported in 1993 by a team of physicists—Charles Bennett of IBM's Watson Research Center; Gilles Brassard, Claude Crepeau, and Richard Josza of the University of Montreal; the Israeli physicist Asher Peres; and William Wootters of Williams College—that rely on quantum entanglement (Chapter 4).

Remember, two entangled particles, say two photons, have a strange and intimate relationship. While each has only a certain probability of spinning one way or another, and while each, when measured, seems to "choose" randomly between the various possibilities, whatever "choice" one makes the other immediately makes too, regardless of their spatial separation. In Chapter 4, we explained that there is no way to use entangled particles to send a message from one location to another faster than the speed of light. If a succession of entangled photons were each measured at widely separated locations, the data collected at either detector would be a random sequence of results (with the overall frequency of spinning one way or another being consistent with the particles' probability waves). The entanglement would become evident only on comparing the two lists of results, and seeing, remarkably, that they were identical. But that comparison requires some kind of ordinary, slower-than-light-speed communication. And since before the